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Challenging Normalized Gendered Racism in Departmental Efforts to Broaden Participation in Computer Science

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In recent decades, initiatives to diversify post-secondary educational spaces have blossomed. Many of these "broadening participation" efforts are in STEM undergraduate departments that, historically and presently, predominantly serve white men. Using a raced-gendered theoretical lens, we conducted a narrative analysis of interviews with 55 faculty, staff, students, and administrators at four public universities, individuals who had all been identified as leaders of diversity, equity, and inclusion efforts in their universities' computer science departments. Our findings across institutions illustrate how their efforts often contravened the stated goals of diversity initiatives, especially for Women of Color, and continued to reproduce white supremacy in computer science. This intersectional inquiry affirms prior scholars' calls to name and confront how diversity initiatives and their leaders deflect to gender while being race-avoidant, a pattern that functions to further marginalize Women of Color in higher education.

For decades, practitioners, policymakers, and researchers have raised concerns about "underrepresentation" within the field of computer science (CS), highlighting persistent numerical disparities and inequities in secondary and postsecondary student enrollment and degree attainment by race, gender, geography, income, and ability (e.g., National Science Foundation, 2011, 2018, 2020a, 2020b; Singh et al., 2007; Varma, 2018). As a discipline, CS has developed in a way that particularly excludes Students of Color from access to a computing education and the tech field (Margolis, 2017). The NSF has strongly supported and funded

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"strategies for broadening participation" to address inequities in the CS educational pipeline by attending to "the most pressing underrepresentation concerns" (National Science Foundation, 2008, p. 3–4) and called on private industry to bolster these efforts with financial investments. Although multiple student groups have been identified as underrepresented in CS education, gender disparities have been consistently elevated "as the central logic in the common discourse of underrepresentation," motivating "a plethora of gender-targeted" efforts to recruit and retain women in recent years (Convertino, 2020, p. 595). Hence, within initiatives to "broaden participation in computing" (BPC) that focus on marginalized groups, gender often maintains the most pressing attention, leaving other equity work, such as addressing and eliminating racial disparities, at the periphery.

In this study, we consider how whiteness shapes the implementation of gender equity work and, specifically, problematize the ways that Women of Color experience initiatives that are nominally built to serve their needs. To do this, we conducted a secondary qualitative analysis of BPC initiatives at four university campuses in the United States to observe, identify, and call out how racism and sexism operate simultaneously through these efforts to shape the experiences of Women of Color. The following research question guided our exploration: What raced-gendered dynamics are embedded in departmental diversity, equity, and inclusion efforts to broaden participation in computing? We applied a critical, raced-gendered lens to our data to elucidate how gendered racism is operationalized within BPC efforts and to validate the narratives and embodied intuition of Women of Color who navigate similar dynamics in CS.

LITERATURE

Introduced by the NSF in 1997, the language of "broadening participation" in STEM fields refers to the United States federal government's desire "to increase the size, talents, and diversity of America's science and engineering workforce" (Colorado State University, 2020, para. 6). As CS fields have long been representationally dominated by white, cisgender men from affluent socioeconomic backgrounds (Margolis, 2017; Nakajima et al., 2022), BPC efforts have focused on shifting the compositional diversity of CS fields by increasing the number of students from "underrepresented" communities—namely, women and "underrepresented minorities" (Colorado State University, 2020, para. 6). Findings from our previous work (Nakajima et al., 2022) align with Bensimon (2016), that this rhetoric of underrepresentation permeates CS education and drives diversity efforts in these fields in a way that collapses meaningful differences among marginalized groups.

Our study extends critiques of how the discourse around underrepresentation is inadequate in conceptualizing how norms and structures reproduce inequity in CS. Said differently, we view the underrepresentation of Women of Color in CS as a symptom of societal systems of gendered racism that permeate postsecondary STEM education; the collapsing of Women of Color under broader language of "underrepresented minorities" similarly mirrors these broader systems of domination (Williams, 2020). Within STEM education, critical scholars have increasingly problematized terms like underrepresented racial minority (often attenuated to URM) for functioning as racist language by: (a) denying "groups the rights to name themselves" (Williams, 2020, para. 11); (b) erasing the complex differences among underrepresented groups; and (c) avoiding engaging with norms and structures that produce overrepresentation.

Scholars have long articulated the problematic byproducts of underrepresentation in CS (Holloman et al., 2021), including the lack of resources for racially minoritized students in CS departments (Braziel & Braziel, 2001; Figueroa, 2015) and the absence of Black women mentors for Black women students (Squires, 2016). However, BPC initiatives continue to be motivated and rationalized by this rhetoric. Although our study does not document the racialized and gendered implications of underrepresentation, we review the multifaceted literature on URM in CS as inextricable from and background to our inquiry. Our review is organized into Williams's (2020) three critiques: "underrepresentation is not a permanent fixture of a group's identity" (Williams, 2020, para. 25); erasure of "differences in the circumstances" (para. 13); and "overrepresented majorities" (para. 15). This is crucial to interpreting our key finding, that BPC initiatives can be rooted in and perpetuate systems of domination they were meant to redress.

"Underrepresentation is Not a Permanent Fixture of a Group's Identity"

The practice of categorizing groups of people as minority and URM is adopted "by those who have the power to produce the numbers [and] shape the reality of racial inequity" (Bensimon, 2016, p. 3). White, economically privileged, entrepreneurial, stereotypical "hacker-types" who dominate the masculine computing fields (Eglash, 2002; Kendall, 2011) use such constructs to signal intellectual superiority and reinforce the racist assumptions that People of Color are underprepared for the rigor and complexities of CS work (Malcom et al., 2005). This attitude has justified a lack of meaningful inclusion efforts that support and welcome People of Color in university computing departments; rather, BPC foci have been on assimilating and recruiting People of Color at younger ages to influence their likelihood to choose CS as a major in their college-going years (Lehman et al., 2021).

Despite this language of underrepresentation centering white people and elevating deficit thinking of individuals who are not identified as white (Williams, 2020), labels such as URM are frequently adopted in both higher education and the sciences to refer to the relatively low numerical or statistical presence of certain populations (Bussmann et al., 2020). The use of URM as an aggregate term—which can be harmful as it abbreviates and trivializes groups of people whose identities were labeled by others (Williams, 2020)—is often embraced by CS equity researchers and practitioners when working to address compositional inequities in CS (e.g., Kulkarni et al., 2018; Sax et al., 2017; Sloughter et al., 2019). Scholars have both theoretically (Williams, 2020) and empirically problematized the use of URM. For example, Yamaguchi and Burge (2019) found that Black women in CS reported that being referred to as "under" (as the term "underrepresented" implies), made them feel minimized and inferior to the (white) majority.

Erasure of "Differences in the Circumstances"

Emphases on underrepresentation in diversity initiatives can also obscure meaningful intersections of identity across and within groups, to further undermine and efface people who experience compound levels of oppression in computing. Consequential and observable differences for Women of Color are often lost (A. Scott et al., 2017) when BPC efforts focus on "women and minorities" as an aggregated category (Gilbert et al., 2015; London et al., 2020;

Trauth et al., 2016). Unfortunately, this language and practice continues in CS education and BPC, particularly as the dominance of white men in high-technology industries and in computing has become starkly apparent (Camp et al., 2017).

Scholars can also ignore overlapping identities by using gender or race (but not both) as a primary organizing framework (Lunn et al., 2021). Even as research teams racially disaggregate the purported gender gap to demonstrate how CS presents unique barriers and challenges for Women of Color, there continues to be silence around how white women might participate in and benefit from the conditions that create these disparate outcomes (Convertino, 2020).

Perhaps as a form of resistance, Women of Color scholars who examine both gender and race in CS are increasingly turning to critical frameworks—like intersectionality (Crenshaw, 1989, 1991)—to document how Women of Color navigate STEM education and professional spaces while simultaneously experiencing the "double bind" of both sexism and racism (Ong et al., 2011, p. 175). Intersectionality has helped to name the many ways that Black women college students are treated as anomalies in computing departments and excluded from different communities for not being men and for not being white (Ashford, 2016; Charleston et al., 2014; Rankin & Thomas, 2020; Sanders, 2020). Specifically, these scholars have attributed the slow progress of BPC efforts to leaders' focus on gender at the expense of other aspects of identity (Yamaguchi & Burge, 2019), and "their failure to acknowledge or appreciate how intersectionality has shaped the lived experiences of women of color navigating the computing ecosystem" (Rankin & Thomas, 2020, p. 199). We build on these critiques by identifying how white women in CS equity work absorb the rhetoric of underrepresentation and of intersectionality to avoid attending to critical questions of power and racial dominance.

"Overrepresented Majorities"

The third consequence of emphasizing underrepresentation in CS diversity work is the convenient side-stepping of the topic of racism and ignoring "the roots of inequality" (Bensimon, 2016, p. 6). Throughout our review of the literature in the above two sections (denying groups to name themselves and "aggregating groups together [to] hide data;" Williams, 2020, para. 13), we demonstrated how Women of Color are treated with hostility and dismissed in CS (Ashford, 2016; Charleston et al., 2014; Rankin & Thomas, 2020; Sanders, 2020). However, as scholars study these symptoms and conditions, we have simultaneously observed a lack of critique of how groups become overrepresented (Williams, 2020), why these differential experiences exist, and who should be held accountable for the multiple marginalizing effects of racial hierarchies (Lunn et al., 2021; Rankin & Thomas, 2020; Sanders, 2020; Yamaguchi & Burge, 2019).

Although women are welcomed into CS departments and teams through BPC efforts, the field remains racially homogenous because of the absence of intentional community-building efforts that center People of Color generally and People of Color with minoritized gender identities (Ross et al., 2020; Yamaguchi & Burge, 2019). As equity-centered scholars in CS, we have experienced these phenomena firsthand and continue to observe how whiteness is elevated on our research teams, in the academic literature, and at professional conferences for CS education practitioners and researchers, even when few white men are present. We voiced our frustrations with like-minded scholars and believe that these dynamics exist by design and are reified by white women in BPC. This study was born out of our desire to unapologetically

name, discuss, and contest these dynamics that contribute to white women being overrepresented in diversity and inclusion efforts, and how they are experienced by Women of Color in CS.

THEORETICAL PERSPECTIVE

In an earlier analysis of BPC efforts, we observed how gender diversity work was prioritized, situating Communities of Color outside of BPC efforts and marginalizing the labor and identities of Women of Color (Nakajima et al., 2022). However, fully examining the latter set of dynamics was beyond the scope of that inquiry. We embarked on this secondary analysis with a commitment to consider the data through a theoretical frame that simultaneously recognized gender and racial oppression rather than gender or race separately, a practice that ignores how the two are mutually informing. In doing so, we embraced what Delgado Bernal (2002) described as a raced-gendered epistemology that takes up "culturally specific ways of positioning between a raced epistemology that omits the influence of gender on knowledge production and a White feminist epistemology that does not account for race" (p. 107). Such frameworks include a range of theoretical traditions like Black feminist thought and Chicana feminist epistemologies. We recognize these multiple genealogies not to efface the Black feminist tradition of intersectionality (Crenshaw, 1989, 1991), but to read these different articulations simultaneously (Nash, 2018) alongside critical race theory (Sampson, 2017; Vaccaro & Camba-Kelsay, 2016), a theoretical lens emerging from Black legal scholarship to explore race and racism as "fundamental organizing principle[s] of social stratification" (Omi & Winant, 2014, p. 107). To do so, we drew from critical race feminism, which integrates and names how gendered racism shapes the experiences of Women of Color and preserves white dominant culture in education and society (Berry, 2010; Wing, 1997).

Broadly, critical race feminism encompasses the "theories and practices that simultaneously study and combat gender and racial oppression [by acknowledging how] Women of Color's experiences, thus perspectives, are different from the experiences of Men of Color and those of white women (Evans-Winters & Esposito, 2010, p. 20), and "disrupt[s] essentialist, monolithic notions of who we are, aspire to be, and can become" (Croom & Patton, 2015, p. 70). Critical race feminism and other raced-gendered epistemologies share methodological approaches with critical race theory through their emphasis on counterstorytelling, subjectivity, situatedness, and the lived experiences of People of Color (Berry, 2010). As a tenet of critical race theory, counterstorytelling can disrupt normalized, racialized discourses of the dominant groups that perpetuate stratification of power by centering Communities of Color (DeCuir & Dixson, 2004; Hiraldo, 2010). Counterstorytelling is often used in critical race feminism to position Women of Color "in the center, rather than the margins of the discussion, reflection, theorizing, research, and praxis of our lives" (Berry, 2010, p. 22).

Our counterstorytelling illustrates the raced-gendered dynamics that Women of Color have to navigate in CS departments and concretize white supremacy under the guise of BPC initiatives. Specifically, we developed an illustrative narrative that exemplifies the tensions we documented during data collection at one institution, and synthesize how these themes

emerged across all four case sites included in this study. Centering Women of Color's experiences and insights in these ways allows us to name how gender is uncritically prioritized within BPC efforts, unpack the devastating influence of such practices on Women of Color, and critique how whiteness is maintained in computing spaces through the rhetoric of underrepresentation.

METHODS

This study of narratives is a secondary analysis of interview data from four institutions expanding racial and gender diversity within their computing departments as part of a national BPC initiative. The impetus to conduct a secondary analysis was, in large part, related to how we, as scholars with marginalized identities, perceived tensions in participants' narratives about the BPC efforts in their department. As such, we open the methods section with our positionality statements before describing our data sources and our approach to this secondary analysis.

Researcher Positionality

Throughout the project, we reflected on how we oriented ourselves to racial and gender equity work individually as scholars and as coauthors, and on our outsider status. Although we served a research institution that primarily studied BPC efforts in colleges and universities, none of us identified as computer scientists nor had a computer science degree. With one another, we shared our own histories, positionalities, and familiarity with raced-gendered perspectives and critical scholarship, being explicit about the theories and assumptions that shaped us and every aspect of this project. D'Silva et al. (2016) recommended examining reflexivity and positionality in this manner, especially for researchers of minoritized communities. In their spirit, we invite our reading audience into this discussion.

Julia: As a mixed, Black and cisgender woman with experience as a student affairs practitioner, I have experienced, and remain motivated to amplify, how racism and sexism are experienced by Women of Color who are staff in higher education. In our early phases of data collection and analysis, it felt clear that, by aggregating data at the department-level, we would be minimizing the insights of the Women of Color we spoke to, who critically identified how BPC efforts are not isolated from the systems of power they purport to disrupt. I am grateful to be partnered with colleagues who recognized the need to center and unpack these dynamics.

Tomoko: I am an US-born woman with Asian heritage. I am motivated to affirm counter-narratives because, like the Women of Color participants who are the focus of this study, all of my post-secondary educational and professional experiences have been in predominantly white spaces where I felt I did not belong. Throughout our analysis, I also recognized the need to deepen my relationship with Black feminist scholarship and critique the ways that I participated in and benefitted from institutionalized anti-Blackness in schooling, where Asian individuals can be perceived as white-adjacent.

Justin: My journey as a scholar-practitioner has been shaped heavily by systems of domination. As a white settler who has exclusively navigated predominantly white institutions, I have long benefitted from the same white supremacy perpetuated in postsecondary

education that this study critiques; at the same time, my experiences as a trans, nonbinary person nuance the ways I understand gendered dynamics in CS. I name the power my whiteness affords me as an author, and acknowledge how my gender colors my understanding of gendered diversity initiatives in CS that exclude my communities by perpetuating the cisgender binary. As I was not involved in data collection, my understanding of the raced-gendered dynamics observed is filtered both through my positionality and the labor of my coauthors.

Data Source

This article is a secondary analysis of interview data that our research team collected in 2019 from universities endeavoring to diversify their undergraduate majors in computing fields. As part of the original study, four institutions were selected as they marked increases in their CS departments' proportion of women and racially underrepresented students. In the present study, we refer to them by our pseudonyms Ash, Elm, Oak, and Palm Universities. All four are public, doctoral degree-granting research universities that vary in institution and department size, geographic location, the ethnoracial compositions of their enrolled students, and the types of diversity initiatives implemented in their computing departments.

TABLE 1
Study Participants

	*White***	*Asian*	*Black*	*Latinx*	*Two or more races*	*No response*
*Women**						
Students (UG & graduate)	5	10	2			
Staff	2	3	3			
Faculty (TT & non-TT)	8				1	1
Dept. chair		1				
Non-CS leaders	4		1			
Subtotals	19	14	6	0	1	1
Men						
Students (UG & graduate)	1					
Staff						
Faculty (TT & non-TT)	5	3		1		1
Dept. chair	1	1			1	
Non-CS leaders						
Subtotals	7	4	0	1	1	1

*Participants were asked to select their gender identity from the following list: gender-queer/nonconforming, man, or woman. We also offered a fill-the-blank "self-identity" option that no one selected.

**Participants were asked to select their race/ethnicity identities from the following list: African American/Black; Native Hawaiian/Pacific Islander; American Indian/Alaska Native; Arab, Middle Eastern, or Persian; East Asian, Southeast Asian, or South Asian; Mexican American/Chicano or Mexican, Central American, or Puerto Rican; or White/Caucasian. There was also a fill-the-blank "other" race/ethnicity category that no one selected. We aggregated these data for the sake of brevity and to protect individuals from identification (e.g., Asian, two or more races). We also used the construct Latinx as a gender-inclusive referent for participants of Latin American descent (Salinas & Lozano, 2019).

Data were collected through semi-structured, hour-long, individual Zoom interviews with 55 people who had all been referred to us as leaders of diversity efforts in computing. Our sample included a wide variety of perspectives from undergraduate and graduate students, staff, faculty, and administrators within CS departments and at the campus level (Table 1).[1]

Two of us—Drs. Karpicz and Nakajima—conducted all but one of the participant interviews. During data collection, they were drawn to the narratives of Women of Color in the sample—likely because they both identify as Women of Color. However, the team's primary inquiry informed an audience of CS educators how exemplary departments maintained their diversity initiatives in light of a spike in student enrollment (Lehman et al., 2021). This earlier publication did not describe who led BPC efforts, the effectiveness of the initiatives, nor how these efforts were experienced. Our present study was born out of the need to elevate Women of Color in that dataset, whose nuanced lived realities did not align well with the aims and structures of the larger study.

Secondary Data Analysis

In this secondary analysis of the same interview data, we strayed from the underlying assumptions that guided the original project. We used a critical frame and different analytical methods (Coyer & Gallo, 2005) to examine how Women of Color experienced BPC efforts in their computing department. Hence, this secondary inquiry can be described as a "more in-depth analysis of an emergent issue or aspect of the data, that was not addressed or was only partially addressed in the primary study" (Heaton, 2008, p. 39), to interrogate the sampling method and interview questions designed specifically for the primary analysis (Hinds et al., 1997).

In the initial study, a larger team of seven researchers had excerpted the interview data into a set of five, broad structural codes (Saldaña, 2016) related to areas of interest based on our knowledge of BPC literature and the needs of our CS educator audience. Those codes were: (a) observations of the enrollment growth in CS; (b) strategies to diversify CS; (c) how diversity initiatives were impacted by the enrollment boom; (d) institutional contexts specific to each site; and (e) future plans for CS diversity efforts. The primary inquiry drew interview excerpts from the first and third codes. This analysis pulled primarily from the second category, which captured data about the diversity efforts in computing and the people who led those initiatives, and included excerpts from all 55 transcripts across the four universities. There was minimal overlap in data.

Narrative Approach

We present this study as a narrative analysis, using field texts (e.g., interview data) to recreate a story as the primary unit of interest (Clandinin & Connelly, 2000). This approach can help dismantle oppressive dominant epistemologies by centering "narratives that are being told by minorities and populations which are underprivileged, [that] may be lost to the majority of people" (Misawa, 2012, p. 241). White women were demographically overrepresented in our study of BPC leaders (Table 1). The collective narratives and raced-gendered lens together

[1] See (Lehman et al., 2021) for more details on our data source. We do not disaggregate the sample by institution to maintain participant confidentiality.

allow us to prioritize the first-hand accounts and experiential knowledge (Delgado & Stefancic, 2012) of Women of Color and "embrace the small numbers-ness" of this group in our sample (Pawley, 2019, p. 368). Our own researcher perspectives, which were enriched through data collection and collaborative analysis, also forged new understandings of the narratives, and contributed to this more contextualized, collective understanding (Clandinin & Connelly, 2000) of the effects of BPC work in postsecondary education.

In reviewing the data for this analysis, we grouped 872 interview excerpts where participants discussed BPC initiatives into categories by initiative type (e.g., women-focused, centering racial equity), noting who led these initiatives and how they were supported within the department. We observed that leaders in gender-focused spaces were all white women and that racial equity initiatives were largely led by Women of Color. The degree of departmental support for these endeavors differed, most conspicuously at Ash University, where white women directed efforts focused on gender diversity from a centralized, well-funded center, as Black leaders organized race-focused grassroots initiatives with little support or recognition from CS department administrators.

Thus, the narratives around raced-gendered dynamics in BPC initiatives included both a dominant narrative—how gender equity was integral to advance departmental BPC efforts—and a counterstory about how Women of Color were not equitably supported by these efforts. A counterstory emerged through the narratives of Women of Color in our sample that highlighted their experiences being denied access to departmental resources and having their participation tokenized within women-focused spaces in CS. Our findings call attention to the antagonistic dynamics that emerged when gender and racial diversity efforts were uncritically collapsed under the umbrella of broadening participation for underrepresented groups.

FINDINGS

We present the themes that emerged from our expansive dataset of excerpts from 55 interviews in two sections. In the first, we introduce the two initiatives at Ash University participants reported as the prominent cocurricular efforts within the department to build community and belonging for minoritized students. Drawing on all the interviews from Ash, we summarize each initiatives' goals and efforts to advance racial and/or gender diversity. The dominant narrative about BPC efforts at Ash is that impactful diversity programs grew in popularity and were institutionalized. The counterstory that follows centers the insights of Women of Color leaders who spoke to these power inequities.

Although this first section of findings is contextually specific to Ash, it illuminates trends in how all four universities engaged in BPC efforts. As such, the second section of our findings integrates narratives from Women of Color across the dataset to echo and name similar dynamics at the other three institutions: empty signals of inclusion and inequitable resource allocation.

Dominant and Counter Narratives About BPC Initiatives at Ash University

Despite the CS department being one of the largest in an institution where the majority of undergraduate students identified as People of Color at the time of data collection, study participants recalled few Faculty of Color in the department over the past several decades. Therefore, two initiatives to BPC were founded: the United Technical Talks (UTT) and the Women's Support Network (WSN). This section outlines the histories of and responses to these initiatives.

Initiative 1: United Technical Talks (UTT)

To address the impact of the CS department's racial disparities on students, three Black staff members employed in the CS undergraduate advising office at Ash founded the UTT in 2017. Their grassroots initiative invited prominent CS Scholars of Color from around the country to present their cutting-edge research at Ash University, with the goal that "underrepresented students" could see "other People of Color in positions of authority and as scholars," Sydney said. Students of Color were particularly encouraged through the UTT experience—thinking, "This person looks like me; I could do this!" (Ray).

The speakers' renown filled every UTT event to full capacity with a racially diverse group of undergraduate and graduate students, faculty, and even people from outside the CS department, to enjoy the "incredible experience" of learning about "new and amazing" developments in technology (Lisa). "It was good for non-Students of Color to see that," Sydney said, adding that she had wanted white students to notice how racially homogenous most computing spaces were, begin to ask questions about "who's not in the room," and demand action. The CS department chair also asked to meet all of the guest speakers to try to recruit them, all prestigious Professors of Color, to Ash University. More than half of our participants mentioned the UTT and its three Black founders as leading efforts to help make CS "feel more welcoming to every single individual student" (Lisa).

The organizers intentionally centered the UTT around the research and scholarship of Faculty of Color, rather than on "the plight of being underrepresented" in CS (Ray). Visiting scholars were encouraged to share how they navigated marginalization in CS privately, in meetings that the organizers arranged with Ash University Students of Color. In these closed spaces, students connected with the guest speakers on a more personal level, developing their professional networks to garner mentors, jobs, and internships at the guests' institutions. Jordan told us that the department rarely promoted opportunities that felt relevant to her and other Black CS majors, but "where I do get good information is with the UTT."

Initiative 2: The Women's Support Network (WSN)

Ash University was also home to the WSN, a college-level administrative office founded in 2014, where efforts to draw "women into the field of computing and further grow the computing pipeline"[2] were centralized. The WSN was led by Dr. S, a professor recognized

[2] We withhold the name and URL of the website to maintain confidentiality.

widely as "the diversity point person for the whole department," (Rhonda) having been the inaugural chair of both the Faculty Committee on Diversity and serving for years as faculty advisor to the all-women CS student organization.

From its founding, Dr. S and the Network office worked to broaden the pipeline of girls interested in CS, by hiring women CS majors to teach computing at nearby K-12 schools and at a middle school summer coding camp "designed to be attractive and accessible to [women] students, [though men] students from underserved populations are encouraged to apply". Through the years, the WSN employed "the most amazing tutors who are also women in CS," hosted social events for the "small number of women in [upper-level undergraduate] classes," and initiated a peer mentoring program to encourage "a really good support system" for women CS majors (Ashley). This office was also in charge of distributing departmental funds earmarked for professional conference attendance, and most of those awards went to women. White women who were employed and/or supported by the WSN felt empowered by the space, with one undergraduate student describing that she was "very proud to say that I'm a part of" the WSN community (Ashley).

Matthew, a white CS professor, marveled, "There's [departmental] support for everything that the WSN does." At the time of our interview, Dr. S had a staff of full-time professional and student workers. Though the WSN had largely focused on the gender disparity and not on other forms of inequity apparent in CS, its success positioned Dr. S at the center of all diversity, equity, and inclusion efforts within the department. When a donor invested significant funding to supplement diversity-related activities in CS, the department's Faculty Committee on Diversity was "redesigned … to be an advisory board for the Inclusion Plan" (Dr. S) and Dr. S was appointed as director of the new "umbrella initiative" (Elaine). In reminiscing about the initial years of the WSN, Dr. S said, "It felt much more grassroots … then it got bigger and bigger. We've reached a point now that we're actually starting to come down from the top, which I think is the real way to have impact."

Identifying Tensions: A Counterstory of the Ash University Initiatives

In reflecting on the department's new Inclusion Plan, Lisa, a UTT founder, noted that there "definitely were some racial dynamics" that preceded and shaped the organization of this new umbrella initiative that "we couldn't, for some reason, talk about." These dynamics centered around the power of Dr. S, the founder of the WSN, within the department, in particular her role in approving department resource disbursements for BPC efforts. According to Lisa and other participants, Dr. S cultivated a reputation of "owning diversity" efforts in CS. In a meeting of the department's Faculty Committee on Diversity, Dr. S even confronted the UTT organizers for "doing diversity wrong," insisting that the guest speakers should address "the very struggles" of being underrepresented in computing rather than only highlight their research. Lisa felt this interaction was condescending, like "a white woman telling three Black people that they didn't understand how diversity should work." The UTT operated "without any funding" in its first semester because, as Ray recalled, Dr. S "thought we were stepping on her toes" and "was able to hold up any money we were trying to get." Unlike the WSN, the UTT initially had "no funding, no support" from the CS department, and the organizers had to seek the support from faculty to invite their colleagues from "top universities" and speak for free (Lisa).

With the umbrella initiative under Dr. S's sole leadership, the WSN was "rebranded" to signal that "everyone," meaning all "underrepresented" students, could now access WSN services. Its name was changed and the "handful of very specific programs just for women" were deemphasized, although a WSN employee admitted that Students of Color were not "explicitly invited to the [WSN] table" to inform these programmatic shifts (Rhonda). Lisa reflected that, "calling something 'Inclusion' didn't mean that it was inclusive," but that this rebrand evidenced a longstanding practice of race being an afterthought. She critiqued: "These white women [at WSN] who came from a very particular background . . . thought, 'We can just add on Minority students to the work that we're doing with women.'" "I feel like inclusion efforts are not taken as seriously," Sydney added, "People can say, 'I'm a part of the Committee on Diversity' and they put that on their resume . . . They have done nothing, no one's held accountable for that."

Behind the scenes, the UTT was subsumed under the Inclusion Plan and under Dr. S's leadership in a multistage process. First, none of the UTT's founders were invited to join the advisory board for the umbrella initiative. Then, the UTT and all student activities (clubs, tutoring, outreach, academic conference attendance, etc.) were brought under the purview of the rebranded WSN, without the input of individuals who originated or led them. Lisa noted that the overt implication was that their "services were no longer needed."

By the time they were interviewed for this project, all three UTT founders had left the CS department and were employed elsewhere. "[I] still feel a little bit of guilt about leaving the students—because those were the people you love working with—and also other staff members and faculty members," Lisa explained, "but [my new job] is giving me the chance to do the work that I couldn't do at Ash University." Jordan, a Black woman undergraduate student at Ash University, expressed sadness about Lisa's departure. "I need a mentor, I need somebody to talk to," she told us. With all the changes happening as a result of the Inclusion Plan, Jordan felt that "Black students are not really involved at all in diversity efforts, . . . that stems from the fact that our voices have been so marginalized."

Themes of Marginalization

The UTT and WSN counterstory demonstrates a motif observed at Elm, Oak, and Palm as well: that gender diversity efforts within BPC initiatives can marginalize Women of Color. We now focus on two subthemes, drawing on our analysis of all 55 interviews at four universities.

Empty Signals of Inclusion

Across institutions, offices and groups originally formed to support women in computing shifted their names to signal broader support for all marginalized students, particularly Students of Color, in computing. In the UTT-WSN example, when the WSN "chang[ed] our model," to begin welcoming students of all minoritized identities, the first step was removing the word "women" from the center's name (Rhonda) to "add on minority students" (Lisa). Similarly, Elm University's student chapter of the Association for Computing Machinery's Council on Women in Computing (ACM-W) rebranded itself as Computer Science for Everyone. However, these actions appeared largely superficial. For example, Elm's rebranded student organization still maintained funding and sponsorship from its international parent organization and its "big

focus on women" (Vanessa). This "new" student organization continued to invest primarily in "opportunities for women to get involved in the department . . . providing them with a support group, a lean-in circle" (Julie).

Women of Color at these institutions expressed skepticism about whether this effort to rebrand the names and missions of women-centered spaces to "be more inclusive of" Students of Color reflected an authentic commitment to racial equity. For example, at Ash University, Lisa felt that the white women organizers who organized the rebranding of the WSN were "not really thinking about the diversity within women, not thinking about trans women, not thinking about Women of Color." Moreover, despite efforts to create space for People of Color within women-supporting initiatives, we observed that many Women of Color students already belonged in, were in the process of building, or had sought support among other communities in and around the department. For example, although Computer Science for Everyone, referenced above, was promoted as "*the* club on campus for diversity and inclusion in computing" (Julie), most Elm University student interviewees noted that Students of Color gravitated toward another organization, Computate.

Computate was an initiative to teach CS at local elementary and secondary campuses after-school settings, led "primarily [by] underrepresented students who are interested in outreach activities" with young children, Denise (a white woman) surmised. However, when we interviewed three undergraduate Women of Color, we were told that they joined Computate for the peer camaraderie and support, as well as the opportunity to develop their technical skills and professional identities through the club's leadership training, mentoring, and community service. Vanessa, a Woman of Color CS major, was attracted to the club's larger scope, "not only women, but underrepresented kids with lower income." She clarified, "What drew me there was not just diversity in gender, but also diversity in skills and in background." Elm had few spaces for people like Vanessa. "We only have two groups: Computate and ACM-W," Vanessa added, "I totally agree that we need more women in tech, but there needs to be a [broader] effort, [or else] you're losing a bigger pool" of interested students, including Women of Color.

Inequitable Resource Allocation

We observed at all four institutions that Women of Color leading or participating in race-centered spaces were often challenged to find funding and material support, while support for gender-focused spaces was celebrated. For example, the WSN-UTT vignette narrates how the white women gained influence over departmental BPC resources were allocated, forcing UTT leaders to operate with little to no support because they were "doing diversity wrong" (Lisa).

Other initiatives centering racial affinity groups at Elm, Oak, and Palm Universities struggled similarly as women's groups expanded their role in departmental diversity efforts. For example, Vanessa remembered her department at Elm University being "very generous" in supporting Computate's activities initially, but as the "years have gone on, [funding has] been cut, it's been getting less and less"; at the same time, their local chapter of ACM-W was expanding with a rebrand as Computer Science for Everyone. Jane, a Woman of Color graduate student, reported that all of the student organizations in CS at Oak University were "only for women. The focus has been on women, but not other underrepresented groups." Their experiences echoed the dynamics at Ash University where, as Lisa noted plainly, "white women have

benefited the absolute most" from departmental BPC initiatives while conversations around intersectionality and the "diversity within women" seemed largely absent.

The CS departments at all four sites also consistently engaged with the Grace Hopper Celebration (GHC), a conference for women technologists, by making it the primary beneficiary of departmental BPC funding. Travel costs for students, faculty, staff, and department chairs were subsidized, classes were canceled during GHC week, and Women of Color were overwhelmingly encouraged to attend GHC. In contrast, such departmental investments and representation were absent from race-centered conferences like Tapia and AfroTech.

At Ash University, when staff and students pushed the WSN to release funds for Black students to attend AfroTech, "There were lots of people who were like . . . 'We've never heard of that. We take our students to Grace Hopper,'" Lisa remembered. Rhonda, a WSN staff person, said that a very recent decision had been made "to not just support Grace Hopper but to support all diversity conferences in computing." Such material gains for race-centered spaces came about only through Students of Color "being visible and making their concerns heard [through] activism and protest" (Lisa). Several Women of Color we spoke with perceived that although spaces like the WSN and GHC were meant "to support all women, that was the purpose, what I've heard from students and Women of Color who are students, [their function] has been, in practice, to support mostly white women" (Sydney).

DISCUSSION

We sought to answer the question: What raced-gendered dynamics are embedded within computing departments' efforts to broaden participation? As others have observed (e.g., London et al., 2020; Sax et al., 2017), we found that departmental efforts to "broaden the participation" of "underrepresented groups" in computing often centered and prioritized gender equity work. This positioned whiteness as "the natural and unmarked racial backdrop for other social positions," by failing to consider racialized power dynamics within women-focused spaces and among BPC leadership (Carbado, 2013, p. 823). Returning to Williams's (2020) critique about the construction of "URM" as a category, we observed several similar dynamics: rebranding the purpose and name of historically women-focused spaces to signal that Women of Color were included without acknowledging the divergent experiences of Women of Color without gendered racism in CS or engaging with the norms and structure that produced an overrepresentation of white women in gender equity work.

For example, the composite narrative from Ash University documented how a white woman professor in CS garnered enough authority to "own" diversity work. When Women of Color in the same department created their own programs for Students of Color, the professor initially dismissed their efforts, then prevented them from accessing departmental resources, and ultimately took over their programs. These dynamics were reflected in interviews with Women of Color across the four institutions in this study. Their counterstories documented how broadening participation efforts in CS—when developed by dominant groups—often alienated Women of Color and hindered their access to institutionalized resources for women.

Women's student organizations were often spotlighted and prioritized within departmental diversity efforts. Women, particularly white women, reportedly had easier access to

departmental resources, like funding for conference attendance and other activities, than did affinity groups for Students of Color. Participants' observations that white women seemed to be centered within equity efforts that were geared toward "all" women corroborate the extant literature on the lack of funding for racial diversity initiatives in STEM disciplines (Braziel & Braziel, 2001; Figueroa, 2015) as well as literature about how white women disproportionately benefit from organizational diversity initiatives in and out of computing sectors (Crenshaw, 2006; Twine, 2018).

Critical race feminism calls attention to the ordinariness and acceptance of racism within everyday life, and emphasizes narratives that center the realities of Women of Color and their experiences with gendered racism (Wing, 1996). Our findings demonstrate how racism can be preserved within BPC efforts that uncritically uplift gender diversity. We observed that when members of dominant groups led diversity efforts without reflecting on their positionality, they decentered and devalued the interests of Students of Color (e.g., Owen, 2009).

Across our sites, Women of Color were further marginalized in initiatives led by white individuals, especially white women, who had likely faced gender oppression in computing departments and whose proximity to power was not questioned when their experiences were essentialized in the two-gender paradigm (Moon & Holling, 2020). White women were over-represented in our sample, and we noted that interviewees rarely acknowledged or reflected on their inherent privilege as they participated in CS, a "white space" (Mascarenhas, 2018, p. 154). We recognize that identity is not formed in a vacuum, and the many social identities an individual holds are neither monolithic nor embodied as mutually exclusive to one another. But by uncritically centering gender in BPC efforts, departmental leaders replicated racial hierarchies and solidified whiteness as an institutional orientation (Truesdell et al., 2017).

Scholars have called attention to the ways that white women access and maintain organizational power (e.g., E. K. Scott, 2005), and critical race and feminist theorists have thoroughly affirmed how "inclusion" efforts in postsecondary education create and reinforce systems that repeatedly benefit white women to a greater extent than People of Color (DeCuir & Dixson, 2004; Ladson-Billings, 1998; McCoy, 2006). Among Women of Color, Black women benefit the least from gender equity work throughout time, and uniquely experience the gendered, racist webs of oppression created through society's interlocking matrices of domination (Collins, 1990). We observed Black women's counterstories throughout the dataset describing how they felt unseen, unwelcomed, and unsupported in computing spaces. The Ash University vignette highlights how departmental leaders disregarded Black women, and the raced-gendered theoretical lens used in this article calls out how these dynamics are purposeful.

The counterstories of Women of Color in computing shared here reveal how departmental foci on broadening participation of underrepresented groups failed to attend to the ways race and racism shape the organization and affect multiply marginalized students.[3] For example, the Women of Color students at Elm and Oak Universities who sought spaces of belonging were all East or South Asian, suggesting a need for counterspaces for Asian women within computing. This finding disrupts how BPC scholars traditionally aggregate Asian and white racial groups as majority groups within the field (Castro & Collins, 2021; Lin, 2021), and points to the

[3] Multiply marginalized refers to the experience of living within intersecting systems of oppression; e.g., Women of Color are multiply marginalized in their simultaneous experience of racism and sexism. We use the term gendered racism (Collins, 1990) interchangeably.

importance of addressing the intersectional oppressions that shape Women of Color's experiences. Our study furthers recent scholarship of how systems of domination are replicated in student-focused spaces in CS, such as mentoring programs, and uniquely impact Women of Color (e.g., Wofford, 2022). When BPC efforts whitewash gender equity by silencing and excluding Communities of Color, it reifies rather than disrupts, history's patterns of inequity.

IMPLICATIONS FOR RESEARCH AND PRAXIS

In this section, we outline how our inquiry might inform liberatory, equity-centered research and practice in postsecondary computing contexts and beyond. First, we share the affordances and limitations of adopting the concept of underrepresentation in our own study. We then highlight the importance of critical frameworks in guiding both research and praxis, and how our approach created space for counternarratives to emerge, stories that presented alternative details of the marginalization of experience in computing departments. Last, drawing from our own experiences, we acknowledge the ongoing challenges and harm that face multiply marginalized scholars and practitioners who engage in this work, and call for our strengthening and unification in efforts to abolish gendered racism from places of learning.

We initially became acquainted with BPC practitioners' focus on representation as a racial panacea when reviewing CS education literature and in the collection and analysis of data for our earlier project. We could see how efforts to recruit and admit more Students of Color concealed the root causes of racial disparities in CS and in postsecondary education, and that positioning minoritized groups as "underrepresented" enabled dominant groups to maintain the social order of computing as a white space. To contrast this, we centered race and gender in quiet and ad hoc ways during this study, an approach we believe can be replicated and extended by researchers and practitioners aiming to interrogate and shift patterns of inequities.

For example, in recruiting BPC leaders for our interviews, the snowball method of identifying participants yielded an overrepresentation of white women. We believe, as Rich (1981) theorized, "so long as we can identify only with white women, we are still connected to that system of objectification and callousness and cruelty called racism" (p. 5). We addressed the demographic disparity by intentionally sampling staff and student leaders, not just administrators and faculty. Two of us were present at nearly all 55 interviews, and our identities as Women of Color researchers undoubtedly helped to create a welcoming space for Participants of Color to voice their raced-gendered observations and recommend other Women of Color to be interviewed. In addition, we used the flexible nature of our semi-structured protocol to encourage Women of Color to elaborate on their experiences, resulting in the rich evidence that substantiated this analysis. Knowing that critically informed inquiry methods, like counterstorytelling and pláticas (Gonzalez, 2021; Gonzalez et al., 2023), can help to demonstrate the sources of inequities, we chose to present these findings by pairing dominant narratives from our data with the othered narratives of Women of Color to privilege the latter, even though Women of Color were far underrepresented in our sample.

Thus, we had an acute awareness of the problem of overrepresentation within our own sample, and adjusted our stances and methods numerous times seeking the counterstory. We were fortunate that a small group of Black women shared their narratives with us—experiences that aligned with one another across campuses and with those of the other Women of Color in

our sample. Anchoring this inquiry on Black women participants' stories best illustrated the gravity and pitfalls of BPC work, and Black women's scholarship around gendered racism provided further language for us to unpack and name those experiences. Had we adopted these approaches a priori in the research design rather than our initial race-neutral sampling strategy, however, we likely would have presented even more divergent perspectives and challenges with gendered racism. For example, due to the absence of participants, our findings did not include the perspectives of Indigenous, American Native, and First Nations women, and we could not interrogate cisnormative conceptions of gender because all of our participants identified within the cisgender binary. In particular, we believe that we have more to learn about the experiences of Latinas and Asian women in computing, whose perspectives were less explored in our study. These are several directions we have identified for leaders of inclusion efforts—and the scholars who study them—to extend our collective work to deepen and nuance our understandings of inequity in CS.

In addition, our findings underscore the importance of BPC leaders and researchers who not only recognize the rhetoric of underrepresentation, but also accept the moral charge to envision beyond demographic parity and actualize more equitable CS departments. Such individuals ought to embody or embrace critical philosophies—such as raced-gendered epistemologies (Delgado Bernal, 2002) and intersectionality (Crenshaw, 1989, 1991)—which enabled us to make sense of the complex and intertwined identities and histories of individuals and groups, of dominant and subjugated classes, and how they play out in computing. As scholars who each live within forms of societal oppression that have previously impacted (and continue to shape) our own educational and professional journeys, hearing, analyzing, and elevating participants' experiences with gendered racism drew on our own vulnerabilities and need for liberation. Cognizant of our positionalities, we constantly reflected both individually and together in every part of this project as we simultaneously name the dynamics that continue to marginalize communities in post-secondary education and upset this status quo.

Presently, funding for gender equity initiatives in CS burgeons (Callahan, 2020), while racial inequities in STEM education writ large continue to be exacerbated (Chen et al., 2022). Efforts that intentionally disrupt white supremacy, gendered racism, and other modes of domination in CS thus hold ever-growing significance. Truly liberatory praxis requires BPC leaders who hold dominant social identities to contend with the privileges they have that prevent them from recognizing disadvantages others face; not doing so renders those "fighting against the evils of 'the man'" unable to recognize when they "become 'the man'" in their own movements toward social justice (Oluo, 2019, p. 59). Perhaps because of this, our study affirms that authentic racial inclusion initiatives are most often grassroots, led and informed by Women of Color (Women of Color in Computing Collaborative, 2018), and counterstories are key to exposing how gendered discourses deflect and obscure the implicit structures of whiteness in computing and elsewhere.

We believe it is especially imperative for BPC leaders to define the problem of underrepresentation, i.e., what systems of oppression produce underrepresentation, and how do multiple, different modes of underrepresentation intersect with each other (Collins, 1990; Crenshaw, 1989). Doing so can help articulate a theory of change for how BPC efforts intervene to transform the status quo. This requires that leaders articulate how they are centering racial equity in the norms, programs, and practices within women-focused spaces, beyond changing the spaces' name to signal inclusion for everyone. Funders supporting BPC

and gender equity efforts in CS should intentionally question how departments pursuing resources for BPC efforts are addressing gendered racism specifically, rather than assuming multiple marginalized student groups will be equitably served by a singular strategy.

It is troubling that so few critically informed individuals are in positions of power in CS education and in CS equity research, as white women and men continue to dominate the narratives that inform those fields. As such, we call on white women leaders of BPC efforts and CS equity scholars to educate themselves, be cognizant of how their positionality shapes their interactions with People of Color, and specifically People of Color with marginalized gender identities, and "take responsibility for their racism" (Smith, 1980, p. 48). Advancing beyond the framework of underrepresentation that currently prevails will enable leaders to invite people held in the margins into the discourse. People external to computing can help facilitate conversations to decenter white women and other leaders from dominant groups and identify the discriminatory norms accepted in so-called equity work that severely and negatively impact People of Color in practice. Elevating the rich insights from these contrasting and diverse perspectives will propel BPC efforts and computing culture to be truly inclusive and emancipatory for all.

CONCLUSION

This study used a critical, raced-gendered lens to unpack entangled dynamics present as computing departments expanded admittance to historically marginalized student groups. In particular, we responded to how forms of marginalization can be collapsed in BPC initiatives, how gender equity retained its primacy in this work, and how white women are empowered and engaged as leaders in ways that Women of Color are not.

The counterstories of Women of Color described how BPC initiatives, such as the WSN, centered womanhood without actually welcoming Women of Color, explicitly or implicitly, into the space. Their narratives illustrated how white women who led gender equity work withheld resources from initiatives led by Women of Color to support Students of Color in CS and subsumed racial equity initiatives within their gender-focused efforts. As a result, Women of Color, both students and staff, were unrecognized and marginalized within gender equity work in CS. These counterstories call attention to latent realities in diversity efforts in computing, educational institutions, and our world at large, that we, as authors and BPC researchers, also witnessed and experienced firsthand. It is imperative that scholars and practitioners address gendered racism within their theories of change, interrogate how systemic domination shapes diversity initiatives, and explicitly disrupt how power manifests within these efforts.

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REFERENCES

Ashford, S. N. (2016). *Our counter-life herstories: The experiences of African American women faculty in U.S. computing education* (6171) [Doctoral dissertation, University of South Florida]. USF Scholar Commons. https://scholarcommons.usf.edu/etd/6171/

Bensimon, E. M. (2016). *The misbegotten URM as a data point*. Center for Urban Education, Rossier School of Education, Rossier School of Education, University of Southern California. https://cue.usc.edu/files/2016/01/Bensimon_The-Misbegotten-URM-as-a-Data-Point.pdf

Berry, T. (2010). Engaged pedagogy and critical race feminism. *Educational Foundations*, *24*(3–4), 19. https://files.eric.ed.gov/fulltext/EJ902670.pdf

Braziel, M. E., & Braziel, W. F. (2001). Factors in decisions of underrepresented minorities to forego science and engineering doctoral study: A pilot study. *Journal of Science Education and Technology*, *10*(3), 273–281. https://doi.org/10.1023/A:1016694701704

Bussmann, J. D., Altamirano, I. M., Hansen, S., Johnson, N. E., & Keer, G. (2020). Science librarianship and social justice: Part one: Foundational concepts. *Issues in Science and Technology Librarianship*, *94*(Spring), 1–11. https://doi.org/10.29173/istl62

Callahan, M. (2020, May 4). Northeastern awards grants to 6 universities to help increase the number of women in computer sciences. *News@northeastern*. https://news.northeastern.edu/2020/05/04/northeasterns-center-for-inclusive-computing-awards-grants-to-six-universities-to-help-them-increase-the-number-of-women-in-computing/

Camp, T., Adrion, W. R., Bizot, B., Davidson, S., Hall, M., Hambrusch, S., Walker, E., & Zweben, S. (2017). Generation CS: The mixed news on diversity and the enrollment surge. *Association for Computing Machinery (ACM) Inroads*, *8*(3), 36–42. https://doi.org/10.1145/3103175

Carbado, D. W. (2013). Colorblind intersectionality. *Signs*, *38*(4), 811–845. https://doi.org/10.1086/669666

Castro, A. R., & Collins, C. S. (2021). Asian American women in STEM in the lab with "White Men Named John". *Science Education*, *105*(1), 33–61. https://doi.org/10.1002/sce.21598

Charleston, L. J., George, P. L., Jackson, J. F., Berhanu, J., & Amechi, M. H. (2014). Navigating underrepresented STEM spaces: Experiences of Black women in US computing science higher education programs who actualize success. *Journal of Diversity in Higher Education*, *7*(3), 166–176. https://doi.org/10.1037/a0036632

Chen, C. Y., Kahanamoku, S. S., Tripati, A., Alegado, R. A., Morris, V. R., Andrade, K., & Hosbey, J. (2022). Systemic racial disparities in funding rates at the national science foundation. *eLife*, *11*(e83071), 1–24. https://doi.org/10.7554/eLife.83071

Clandinin, D. J., & Connelly, F. M. (2000). *Narrative inquiry: Experience and story in qualitative research*. Jossey-Bass.

Collins, P. H. (1990). *Black feminist thought: Knowledge, consciousness and the politics of empowerment*. Routledge.

Colorado State University. (2020, February 10). *A history of broadening participation within the national science foundation*. Colorado State University, STEM Center. https://stem.colostate.edu/a-history-of-broadening-participation-within-the-national-science-foundation/

Convertino, C. (2020). Nuancing the discourse of underrepresentation: A feminist post-structural analysis of gender inequality in computer science education in the US. *Gender and Education*, *32*(5), 594–607. https://doi.org/10.1080/09540253.2019.1632417

Coyer, S. M., & Gallo, A. M. (2005). Secondary analysis of data. *Journal of Pediatric Health Care*, *19*(1), 60–63. https://doi.org/10.1016/j.pedhc.2004.10.003

Crenshaw, K. (1989). Demarginalizing the intersection of race and sex: A Black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. *University of Chicago Legal Forum*, *1*(8), 139–167. http://chicagounbound.uchicago.edu/uclf/vol1989/iss1/8

Crenshaw, K. (1991). Mapping the margins: Intersectionality, identity politics and violence against women in color. *Stanford Law Review*, *43*(6), 1241–1299. https://doi.org/10.2307/1229039

Crenshaw, K. W. (2006). Framing affirmative action. *Michigan Law Review First Impressions*, *105*, 123–133. http://repository.law.umich.edu/mlr_fi/vol105/iss1/4

Croom, N. N., & Patton, L. D. (2015). Using endarkened and critical race feminist perspectives to question and analyze knowledge production narratives. In F. A. Bonner, A. F. Marbley, F. Tuitt, P. A. Robinson, R. M. Banda, & R. L. Hughes (Eds.), *Black faculty in the academy: Narratives for negotiating identity and achieving career success* (pp. 67–78). Routledge.

D'Silva, M. U., Smith, S. E., Della, L. J., Potter, D. A., Rajack-Talley, T. A., & Best, L. (2016). Reflexivity and positionality in researching African-American communities: Lessons from the field. *Intercultural Communication Studies*, *XXV*(1), 94–109. https://www-s3-live.kent.edu/s3fs-root/s3fs-public/file/DSILVA-SMITH-DELLA.pdf

DeCuir, J. T., & Dixson, A. D. (2004). "So when it comes out, they aren't that surprised that it is there": Using critical race theory as a tool of analysis of race and racism in education. *Educational Researcher*, *33*(5), 26–31. https://doi.org/10.3102/0013189X033005026

Delgado Bernal, D. (2002). Critical race theory, latino critical theory, and critical raced-gendered epistemologies: Recognizing students of color as holders and creators of knowledge. *Qualitative Inquiry*, *8*(1), 105–126. https://doi.org/10.1177/107780040200800107

Delgado, R., & Stefancic, J. (2012). *Critical race theory: An introduction* (2nd ed.). New York University Press.

Eglash, R. (2002). Race, sex, and nerds: From black geeks to Asian American hipsters. *Social Text*, *20*(2), 49–64. https://doi.org/10.1215/01642472-20-2_71-49

Evans-Winters, V., & Esposito, J. (2010). Other people's daughters: Critical Race Feminism and Black girls' education. *Educational Foundations*, *24*(1–2), 11–24. https://files.eric.ed.gov/fulltext/EJ885912.pdf

Figueroa, T. (2015). *Underrepresented racial/ethnic minority graduate student experiences in science, technology, engineering, and mathematics (STEM) disciplines: A cross-institutional analysis of their experiences* [Doctoral dissertation, UCLA]. ProQuest.

Gilbert, J. E., Jackson, J. F., Dillon, E. C., Jr., & Charleston, L. J. (2015). African Americans in the US computing sciences workforce. *Communications of the ACM*, *58*(7), 35–38. https://doi.org/10.1145/2770929

Gonzalez, A., Orozco, R. C., & Gonzalez, S. A. (2023). Joteando y mariconadas: Theorizing queer pláticas for queer and/or trans Latinx/a/o research. *International Journal of Qualitative Studies in Education*. *36*(9), 1741–1756. Advance online publication. https://doi.org/10.1080/09518398.2023.2181433

Gonzalez, S. A. (2021). *Jotería identity and consciousness: Pláticas of co-creation with undergraduate queer Latinx students*. Rutgers Graduate School of Education, Proctor Institute. https://proctor.gse.rutgers.edu/sites/default/files/Joteria%20Brief_0.pdf

Heaton, J. (2008). Secondary analysis of qualitative data: An overview. *Historical Social Research*, *33*(3), 33–45. https://doi.org/10.12759/hsr.33.2008.3.33-45

Hinds, P. S., Vogel, R. J., & Clarke-Steffen, L. (1997). The possibilities and pitfalls of doing a secondary analysis of a qualitative data set. *Qualitative Health Research*, *7*(3), 408–424. https://doi.org/10.1177/104973239700700306

Hiraldo, P. (2010). The role of critical race theory in higher education. *The Vermont Connection*, *31*(1), 53–59. https://www.uvm.edu/~vtconn/v31/Hiraldo.pdf

Holloman, T. K., London, J., Lee, W. C., Pee, C. M., Hawkins Ash, C., & Watford, B. (2021). Underrepresented and overlooked: Insights from a systemic literature review about Black graduate students in engineering and computer science. *International Journal of Engineering Education*, *37*(2), 497–511. https://par.nsf.gov/biblio/10221067-underrepresented-overlooked-insights-from-systematic-literature-review-about-black-graduate-students-engineering-computer-science

Kendall, L. (2011). "White and nerdy": Computers, race, and the nerd stereotype. *The Journal of Popular Culture*, *44*(3), 505–524. https://doi.org/10.1111/j.1540-5931.2011.00846.x

Kulkarni, A., Yoon, I., Pennings, P. S., Okada, K., & Domingo, C. (2018). Promoting diversity in computing. In *ITiCSE 2018: Proceedings of the 23rd Annual ACM Conference on Innovation and Technology in Computer Science Education* (pp. 236–241). Association for Computing Machinery. https://doi.org/10.1145/3197091.3197145

Ladson-Billings, G. (1998). Just what is critical race theory and what's it doing in a nice field like education? *International Journal of Qualitative Studies in Education*, *11*(1), 7–24. https://doi.org/10.1080/095183998236863

Lehman, K. J., Karpicz, J. R., Rozhenkova, V., Harris, J., & Nakajima, T. M. (2021). Growing enrollments require us to do more: Perspectives on broadening participation during an undergraduate computing enrollment boom. In

SIGCSE '21: Proceedings of the 52nd ACM Technical Symposium on Computer Science Education (pp. 809–815). Association for Computing Machinery. https://doi.org/10.1145/3408877.3432370

Lin, S. C. (2021). *"The modern mulan of tech": A critical analysis of Asian American Women And their experiences in computer science* [Master's thesis, University of Illinois Urbana-Champaign]. http://hdl.handle.net/2142/110537

London, J. S., Lee, W. C., Phillips, C., Van Epps, A. S., & Watford, B. A. (2020). A systematic mapping of scholarship on broadening participation of African Americans in engineering and computer science. *Journal of Women and Minorities in Science and Engineering*, *26*(3), 199–243. https://doi.org/10.1615/JWomenMinorScienEng.2020027889

Lunn, S., Zahedi, L., Ross, M., & Ohland, M. (2021). Exploration of intersectionality and computer science demographics: Understanding the historical context of shifts in participation. *ACM Transactions on Computing Education (TOCE)*, *21*(2), 1–30. https://doi.org/10.1145/3445985

Malcom, S., Teich, A. H., Jesse, J. K., Campbell, L. A., Babco, E. L., & Bell, N. E. (2005). *Preparing women and minorities for the it workforce: The role of nontraditional educational pathways*. American Association for the Advancement of Science. https://www.aaas.org/sites/default/files/Complete_book.pdf

Margolis, J. (2017). *Stuck in the shallow end: Education, race, and computing* [updated edition]. MIT Press.

Mascarenhas, M. (2018). White space and dark matter: Prying open the black box of STS. *Science, Technology, & Human Values*, *43*(2), 151–170. https://doi.org/10.1177/0162243918754672

McCoy, D. L. (2006). *Entering the academy: Exploring the socialization experiences of African American male faculty* (Order No. 3208182) [Doctoral dissertation, Louisiana State University]. ProQuest. https://repository.lsu.edu/gradschool_dissertations/3297

Misawa, M. (2012). Social justice narrative inquiry: A Queer Crit perspective. In *Adult Education Research Conference* (pp. 239–246). New Prairie Press. https://newprairiepress.org/aerc/2012/papers/34

Moon, D. G., & Holling, M. A. (2020). "White supremacy in heels": (White) feminism, white supremacy, and discursive violence. *Communication and Critical/Cultural Studies*, *17*(2), 253–260. https://doi.org/10.1080/14791420.2020.1770819

Nakajima, T. M., Karpicz, J. R., & Gutzwa, J. A. (2022). "Why isn't this space more inclusive?": Marginalization of racial equity work in undergraduate computing departments. *Journal of Diversity in Higher Education*. Advance online publication. https://doi.org/10.1037/dhe0000383

Nash, J. C. (2018). *Black feminism reimagined: After intersectionality.* Duke University Press.

National Science Foundation. (2008). *Broadening participation at the national science foundation: A framework for action*. https://www.nsf.gov/od/broadeningparticipation/nsf_frameworkforaction_0808.pdf

National Science Foundation. (2018). *Which groups qualify as "underrepresented" in computing and closely related disciplines?* https://www.nsf.gov/pubs/2018/nsf18101/nsf18101.jsp#q3

National Science Foundation. (2020a). *Broadening participation in computing*. https://www.nsf.gov/cise/bpc/

National Science Foundation. (2020b). *Which groups qualify as "underrepresented" in computing and closely related disciplines?* https://www.nsf.gov/pubs/2020/nsf20110/nsf20110.pdf

Oluo, I. (2019). *So you want to talk about race*. Blackstone Publishing.

Omi, M., & Winant, H. (2014). *Racial formation in the United States* (3rd ed.). Routledge.

Ong, M., Wright, C., Espinosa, L. L., & Orfield, G. (2011). Inside the double bind: A synthesis of empirical research on undergraduate and graduate women of color in science, technology, engineering, and mathematics. *Harvard Educational Review*, *81*(2), 172–208. https://doi.org/10.17763/haer.81.2.t022245n7x4752v2

Owen, D. S. (2009). Privileged social identities and diversity leadership in higher education. *The Review of Higher Education*, *32*(2), 185–207. https://doi.org/10.1353/rhe.0.0048

Pawley, A. L. (2019). Shift the default in "broadening participation" in STEM equity research. *International Journal of Gender, Science and Technology*, *11*(3), 365–373. https://genderandset.open.ac.uk/index.php/genderandset/article/view/668

Rankin, Y. A., & Thomas, J. O. (2020). The intersectional experiences of Black women in computing. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (pp. 199–205). Association for Computing Machinery. https://doi.org/10.1145/3328778.3366873

Rich, A. (1981). Disobedience is what NWSA is potentially about. *Women's Studies Quarterly*, *9*(3), 4–6. https://academicworks.cuny.edu/wsq/496/

Ross, M., Hazari, Z., Sonnert, G., & Sadler, P. (2020). The intersection of being Black and being a woman: Examining the effects of social computing relationships on computer science career choice. *ACM Transactions on Computing Education*, *20*(2), 2–15. https://doi.org/10.1145/3377426

Saldaña, J. (2016). *The coding manual for qualitative researchers*. Sage.

Salinas, C., Jr., & Lozano, A. (2019). Mapping and recontextualizing the evolution of the term latinx: An environmental scanning in higher education. *Journal of Latinos and Education*, *18*(4), 302–315. https://doi.org/10.1080/15348431.2017.1390464

Sampson, C. R. (2017). So it "became white activists fighting for integration?" community organizations, intersectional identities, and education reform. *The Urban Review*, *49*(1), 72–95. https://doi.org/10.1007/s11256-016-0382-9

Sanders, Y. (2020). *Voices of African American women in computer science: Implications for K-12 STEM education and beyond (3825)* [Doctoral dissertation, Loyola University Chicago]. eCommons. https://ecommons.luc.edu/luc_diss/3825

Sax, L. J., Zimmerman, H. B., Blaney, J. M., Toven-Lindsey, B., & Lehman, K. (2017). Diversifying undergraduate computer science: The role of department chairs in promoting gender and racial diversity. *Journal of Women and Minorities in Science and Engineering*, *23*(2), 101–119. https://doi.org/10.1615/JWomenMinorScienEng.2017017221

Scott, A., Martin, A., McAlear, F., & Koshy, S. (2017, June). Broadening participation in computing: Examining experiences of girls of color. In *Proceedings of the 2017 ACM Conference on Innovation and Technology in Computer Science Education* (pp. 252–256). https://doi.org/10.1145//3059009.3059054

Scott, E. K. (2005). Beyond tokenism: The making of racially diverse feminist organizations. *Social Problems*, *52*(2), 232–254. https://doi.org/10.1525/sp.2005.52.2.232

Singh, K., Allen, K. R., Scheckler, R., & Darlington, L. (2007). Women in computer-related majors: A critical synthesis of research and theory from 1994 to 2005. *Review of Education Research*, *77*(4), 500–533. https://doi.org/10.3102/0034654307309919

Sloughter, J. M., Miguel, A., Rempe, M., & Kuder, K. (2019, April). *Survey analysis of student experiences for underrepresented populations in engineering and computer science*. 2019 CoNECD - The Collaborative Network for Engineering and Computing Diversity. https://peer.asee.org/31794

Smith, B. (1980). Racism and women's studies. *Frontiers: A Journal of Women Studies*, *5*(1), 48–49. https://doi.org/10.2307/3346304

Squires, S. M. (2016). *A study of the lived experiences of African American Women STEM doctoral degree completers*. (3689210) [Doctoral Dissertation]. Hampton University. ProQuest.

Trauth, E. M., Cain, C. C., Joshi, K. D., Kvasny, L., & Booth, K. M. (2016). The influence of gender-ethnic intersectionality on gender stereotypes about it skills and knowledge. *ACM Special Interest Group on Management Information Systems Database: The DATABASE for Advances in Information Systems*, *47*(3), 9–39. https://doi.org/10.1145/2980783.2980785

Truesdell, N., Carr, J., & Orr, C. M. (2017). The role of Combahee in anti-diversity work. *Souls*, *19*(3), 359–376. https://doi.org/10.1080/10999949.2017.1389632

Twine, F. W. (2018). Technology's invisible women: Black geek girls in Silicon Valley and the failure of diversity initiatives. *International Journal of Critical Diversity Studies*, *1*(1), 58–79. https://doi.org/10.13169/intecritdivestud.1.1.0058

Vaccaro, A., & Camba-Kelsay, M. J. (2016). *Centering women of color in academic counterspaces: A critical race analysis of teaching, learning, and classroom dynamics*. Rowman & Littlefield.

Varma, R. (2018). U.S. science and engineering workforce: Underrepresentation of women and minorities. *American Behavioral Scientist*, *62*(5), 692–697. https://doi.org/10.1177/0002764218768847

Williams, T. L. (2020). *"Underrepresented minority" considered harmful, racist language*. Communications of the ACM. https://cacm.acm.org/blogs/blog-cacm/245710-underrepresented-minority-considered-harmful-racist-language/fulltext

Wing, A. K. (1996). Critical race feminism and the international human rights of women in Bosnia, Palestine, and South Africa: Issues for LatCrit theory. *University of Miami Inter-American Law Review*, *28*(2), 337–360. https://repository.law.miami.edu/umialr/vol28/iss2/10/

Wing, A. K. (Ed.). (1997). *Critical race feminism: A reader*. NYU Press.

Wofford, A. M. (2022). Equity-minded stage-ahead mentoring: Exploring graduate students' narratives as mentors to undergraduates in STEMM. *The Review of Higher Education*, *46*(2), 249–279. https://doi.org/10.1353/rhe.2022.0017

Women of Color in Computing Collaborative. (2018). *Data brief: Women and girls of color in computing*. https://www.wocincomputing.org/wp-content/uploads/2018/08/WOCinComputingDataBrief.pdf

Yamaguchi, R., & Burge, J. D. (2019). Intersectionality in the narratives of Black women in computing through the education and workforce pipeline. *Journal for Multicultural Education*, *13*(3), 215–235. https://doi.org/10.1108/JME-07-2018-0042